

GeoNatureAgent Benchmark: Benchmarking LLM Agents for Environmental Geospatial Analysis Across Frontier and Open-Weight Foundation Models [Benchmark]

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Abstract

Environmental scientists spend disproportionate effort on data wrangling rather than analysis. New AI agents can be a helpful tool, but their use still requires validation. Furthermore, no benchmark exists to evaluate AI agents that automate environmental geospatial workflows through structured tool calling against real APIs. We introduce the **GeoNatureAgent Benchmark**, the first benchmark for environmental analysis agents that operate via structured tool calls to a production-style geospatial API. The benchmark comprises 93 tasks across 18 categories, covering municipality-level analysis, multi-turn conversation, spatial reasoning, cross-indicator synthesis, error handling and recovery, ranking, comparison, multi-lingual understanding, habitat analysis, deep-dive profiling, temporal change, and task rejection. Tasks are evaluated against an open, self-hostable geospatial API that serves three environmental indicators across Spain and Portugal via sixteen tools (twelve domain-specific operations and four auxiliary). We evaluate nine large language models (Claude Sonnet 4, DeepSeek V3.2, GLM-5, Gemini 2.5 Pro, GPT-4o, Qwen3-235B, GPT-OSS-120B, Llama 4 Scout, and Gemma-3-27B) across Vertex AI, OpenRouter, and Anthropic platforms under three temperature-1.0 seeds per model, reporting capability and per-case cost as orthogonal axes. Results manifest that: (1) Claude Sonnet 4 achieves the highest capability at $60.8\% \pm 0.8\%$, followed closely by DeepSeek V3.2 at $56.3\% \pm 3.1\%$, while no other model exceeds 51%; (2) the cost-accuracy Pareto frontier is occupied mostly by open-weight models (Llama 4 Scout $\rightarrow$ Qwen3-235B $\rightarrow$ DeepSeek V3.2 $\rightarrow$ Claude Sonnet 4), with DeepSeek V3.2 offering 93% of Claude’s capability at $11.6\times$ lower cost (\$0.011/case); and (3) structured tool calling against a real API provides a more discriminative measure of real-world agent capability, with mean accuracies 25–35 percentage points below those reported on general-purpose GIS benchmarks. We further demonstrate geographic and domain extensibility by integrating BigEarthNet V2 land-cover data for Portugal alongside Spanish CO₂ suitability and gully erosion

indicators. GeoNatureAgent Benchmark, the evaluation harness, and the self-hostable API are publicly available¹.

CCS Concepts

• **Computing methodologies** $\rightarrow$ **Neural networks**; *Intelligent agents*; *Multi-agent systems*; *Information extraction*; • **General and reference** $\rightarrow$ **Evaluation**; • **Information systems** $\rightarrow$ **Geographic information systems**; • **Applied computing** $\rightarrow$ *Earth and atmospheric sciences*.

Keywords

Geospatial AI, Benchmark, LLM Agents, Tool Calling, Environmental Analysis

1 Introduction

Environmental monitoring at scale requires multi-temporal geospatial analysis, including the detection of land cover change, the assessment of erosion risk, or the computation of vegetation indices; furthermore, the generation of statistical summaries for regulatory compliance. Practitioners routinely spend the majority of their effort on data wrangling, discovering datasets, reprojecting coordinate systems, and debugging code against remote sensing APIs [17, 32]. This expertise barrier limits the adoption of geospatial methods by domain scientists who understand environmental systems but lack GIS programming skills.

Recent advances in large language models (LLMs) have led to the development of AI agent systems that translate natural language into executable geospatial operations. Architectures range from LLMs with tool pools [1, 32] to fine-tuned specialists [30, 31] to multi-agent systems achieving 85–97% accuracy [16, 18]. Major industry players are investing heavily: Google’s Earth AI integrates Gemini with AlphaEarth Foundations [10], the Planet-Anthropic partnership applies Claude to daily satellite imagery [22], and NASA/IBM’s Prithvi-EO-2.0 provides pre-trained vision transformers [26].

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¹https://github.com/gabrielireland/GeoNatureAgent_Benchmark

However, a critical evaluation gap persists. Public benchmarks for LLM agents in geospatial settings remain scarce, and the few that exist target general GIS tasks (GeoBenchX [14], ThinkGeo [23]) rather than environmental science workflows. Environmental knowledge benchmarks (EnviroExam [11]) evaluate knowledge, not agent behavior. Code-generation benchmarks (UnivEARTH [12]) do not reflect the structured API interaction that production systems employ—58% of LLM-generated Earth Engine code fails to execute. At the foundation model level, GEO-Bench [15] and GEO-Bench-2 [24] evaluate pixel-level vision models, not the agent-level tool orchestration that determines real-world usability.

This work fills this gap through four main contributions:

- (1) **GeoNatureAgent Benchmark**—the first benchmark for environmental analysis agents using structured tool calling against a production-style geospatial API, comprising 93 tasks across 18 categories with a sixteen-tool agent interface, evaluated by eight mechanistic checks per case (no LLM-as-judge).
- (2) **Multi-seed cross-platform evaluation**—a systematic comparison of nine LLMs spanning closed-source models (Claude Sonnet 4, Gemini 2.5 Pro, GPT-4o) and open-weight families (DeepSeek V3.2, GLM-5, Qwen3-235B, GPT-OSS-120B, Llama 4 Scout, Gemma-3-27B), each evaluated under three temperature-1.0 seeds on identical infrastructure with mean $\pm$ -standard-deviation reporting; capability and per-case cost are reported as orthogonal axes rather than collapsed into a single gated score.
- (3) **Cost-efficiency Pareto analysis**—quantitative cost-accuracy mapping revealing that the cost-accuracy Pareto frontier is occupied mostly by open-weight models, with DeepSeek V3.2 offering 93% of Claude’s capability at 11.6 $\times$ lower cost.
- (4) **Reproducible MLOps pipeline and open API**—a fully automated Cloud Build pipeline that produces conversation-trace-level results in object storage, plus an Apache-2.0 self-hostable FastAPI service so the entire evaluation can be reproduced without any private endpoint.

The remainder of this paper is organized as follows: Section 2 reviews related work in geospatial AI agents and benchmarks. Section 3 describes the GeoNatureAgent Benchmark design, the ReAct-style agent architecture, and the evaluation protocol. Section 4 presents the experimental results, including the cost-accuracy Pareto analysis of the nine evaluated models. Section 5 provides a discussion comparing the findings with prior benchmarks and exploring domain extensibility. Finally, Section 6 summarizes the conclusions and suggests future work.

2 Related Work

2.1 Geospatial AI Agent Architectures

The literature reveals four paradigms of geospatial AI agent architectures. The **LLM + Tool Pool** approach pairs general-purpose LLMs with callable tools: GeoGPT [32] uses GPT-3.5-turbo with LangChain, LLM-Geo [17] uses GPT-4 for DAG-based decomposition, and GIS Copilot [1] integrates LLMs into QGIS. **Fine-tuned specialists** sacrifice generality for expertise: GTChain [30] fine-tuned LLaMA-2-7B, achieving 32.5% higher accuracy than GPT-4; EnvGPT [31] fine-tuned an 8B model on 100M environmental

tokens, rivaling GPT-4o-mini. **Multi-agent systems** decompose tasks: GeoJSON Agents [18] achieved 97% with GPT-4o Planner-Worker architecture vs. 49% with a single-agent; GeoLLM-Squad [16] gained 17% from specialized sub-agents. **Cost-efficient hybrids**: Geo-OLM [25] enables sub-7B models to perform within 10% of GPT-4o at 100 $\times$ lower cost.

2.2 Benchmarks and Evaluation

Agent tool-use is evaluated by GeoBenchX [14], ThinkGeo [23] (486 remote sensing tasks), and UnivEARTH [12] (only 33% accuracy on Earth Engine code generation and a 58% code failure rate). GeoBenchX is the closest comparable work to GeoNatureAgent Benchmark: 202 tasks across five categories (merge-visualize, process-merge-visualize, spatial operations, heatmaps/contour lines, and control questions), evaluated via 24 general-purpose GIS tools (data loading, filtering, spatial joins, and choropleth rendering) with an LLM-as-judge scoring protocol (3-point scale, three independent judges achieving 88–96% agreement with human annotations). Top performers are o4-mini and Claude 3.5 Sonnet. Crucially, GeoBenchX tools are low-level primitives operating on local files—load_data, filter_categorical, create_buffer, make_choropleth_map—whereas GeoNatureAgent Benchmark tools are high-level domain operations against a production cloud API.

Map reasoning: MapEval [9] (no model exceeds 67%), MapQA [3] (humans 91%, models < 50%). Spatial reasoning: GPSBench [27], SpatiaLab [28] (humans 88%, best model 55%). Foundation models: GEO-Bench [15] and GEO-Bench-2 [24] evaluate *pixel-level* classification/segmentation across 19 datasets, and PANGAEA [19] finds that foundation models do not consistently outperform supervised baselines.

Key distinction: GEO-Bench evaluates pixel-level model accuracy (*can this model classify a satellite tile?*); GeoNatureAgent Benchmark evaluates agent-level tool orchestration (*can this agent reason about which tools to call?*). Both are complementary.

2.3 Environmental Science AI

Specialized models such as EnvGPT [31] demonstrate that focused fine-tuning on environmental tokens allows smaller architectures to rival general-purpose models like GPT-4o-mini [31]. In the industrial and research sectors, Google’s Earth AI [10] has achieved 82% accuracy on geospatial Q&A tasks (vs 50% baseline Gemini), while MCP-grounded interfaces developed by Pan et al. [21] have reached a factual precision of 4.78/5.0 in air quality monitoring [21]. This technical evolution is increasingly essential to meet new regulatory mandates, such as Spain’s RD 214/2025 [13], which requires approximately 4,000 organizations to report their carbon footprints, thereby driving a significant demand for automated, multi-temporal geospatial analysis.

2.4 The Missing Benchmark

No existing benchmark evaluates environmental analysis agents performing structured tool calling against a real geospatial API with multi-turn conversation, multilingual inputs, and cross-indicator synthesis. GeoNatureAgent Benchmark fills this gap.

3 GeoNatureAgent Benchmark

3.1 Benchmark Design

GeoNatureAgent Benchmark comprises 93 tasks organized into 18 categories (see Table 1), spanning easy (19%), medium (45%), and hard (36%) difficulty levels.

Table 1: GeoNatureAgent Benchmark v5 task categories (93 tasks, 18 categories).

Category	Tasks	Tests
Tool selection	21	Correct tool choice
Cross-indicator	8	CO ₂ × erosion × land cover synthesis
Deep dive	6	Multi-tool environmental profiling
Interpretation	7	Reasoning over analysis results
Error handling	6	Non-existent entities, invalid indicators
Habitat analysis	7	BigEarthNet V2 land cover (Portugal)
Language	6	Galician, Basque inputs
Municipality	4	Municipality-level analysis, disambiguation
Memory (multi-turn)	6	Follow-up questions, context retention
Spatial reasoning	4	Coastal, meseta, community knowledge
Multi-muni ranking	3	Multi-municipality comparisons
Threshold	3	Numeric threshold queries
Temporal change	1	Cross-country temporal context
Error recovery	3	Graceful fallback when data is unavailable
Ranking	2	Top-N and bottom-N province queries
Comparison	2	Close-value and directional comparisons
Single analysis	2	Basic single-province analysis
Province aggregation	2	Aggregate statistics across provinces

Each task specifies: a natural language query (optionally with multi-turn history), expected tool calls, must-contain/must-not-contain strings, maximum rounds, cost budget, and domain-expert ground truth. Error handling tasks are deliberately unsolvable — requesting analysis over non-existent municipalities, fabricating statistics from display-only layers, or citing indicators that do not exist — testing the agent’s ability to decline gracefully rather than hallucinate.

Worked example. To make the format concrete, consider case V5_21 (comparison): “Compare CO₂ geological storage suitability between Murcia and Córdoba. Which province is better suited?” The gold solution issues a single `compare_areas` call over the CO₂ indicator, which returns Murcia 68.5% vs. Córdoba 68.4%; the expected

answer reports both values and notes they are within tolerance ($\Delta 0.1$ pp against a 2.0 pp tolerance), i.e. effectively tied. Every model fails this case, but the failure is instructive: even Gemini 2.5 Pro fails with a *tool-missing* error—it never issues the expected comparison call, so the two comparable values are never produced. The gap is therefore one of *tool selection*, not of distinguishing the 0.1-point margin.

The benchmark covers three environmental indicators across two countries:

- **CO₂ absorption suitability** (categorical, Spain): legislative pre-screening encoding five MITECO spatial criteria [13], producing three classes (Not Eligible, Eligible with Conditions, Eligible). Served as a Cloud Optimized GeoTIFF (COG).
- **Gully erosion probability** (continuous, Europe): machine learning prediction from the LUCAS 2022 soil survey [5] (0–100%). Served as a COG.
- **BigEarthNet V2 land cover** (categorical, Portugal): 7-class land cover distribution derived from 75k+ labeled Sentinel-2 patches [6], aggregated by Portuguese district. Served as pre-computed JSON statistics.

Seven additional layers are available for visual display but not for statistical analysis, creating natural hallucination traps.

To enable direct comparison with general-purpose GIS benchmarks, we integrate four canonical GIS operations from GeoBenchX [14] into our toolset: proximity buffering (`create_buffer`), feature selection by spatial relationship (`select_features_by_spatial_relationship`), centroid extraction (`get_centroids`), and explicit task rejection (`reject_task`) to prevent hallucination on out-of-scope queries. These tools are adapted from GeoBenchX’s local-file paradigm to operate against GeoNatureAgent Benchmark’s cloud API and administrative boundary system. These four operations are available to the agent across the full 93-case suite and are invoked where a query calls for a spatial operation, rather than through a separate set of dedicated cases.

Design rationale. The category structure is a capability taxonomy rather than an arbitrary partition: each category isolates one agentic competency required for environmental geospatial analysis—tool selection, spatial reasoning, multi-turn memory, error handling and recovery, cross-indicator synthesis, interpretation, and multi-lingual understanding, among others. Case counts are weighted by capability importance and coverage rather than balanced: tool selection receives the most cases (21) because choosing the right operation from a realistic toolset is the pivotal agentic skill and the most discriminative across models (Section 4.7), whereas niche capabilities such as cross-country temporal context receive one or two. The uneven sizes are therefore deliberate; we treat the aggregate leaderboard as the primary measure and category-level results as indicative (Section 4). Difficulty labels (easy/medium/hard) are author-assigned by expected tool-chain depth and are approximate.

Tool coverage across data domains. The benchmark’s tools vary in geographic coverage: the CO₂ and gully-erosion indicators (raster COGs, Spain/Europe) are served by the full toolset, whereas the BigEarthNet land-cover indicator (pre-computed district statistics, Portugal) is served only by `analyze_area`. Comparison, ranking, and legend queries over the Portuguese layer therefore require the

agent to compose multiple `analyze_area` calls rather than invoke a single convenience tool. We treat this as part of the evaluation: a capable agent must recognize when a nominally-applicable tool does not cover a layer and adapt accordingly—a realistic tool-selection challenge—and extending the convenience tools to precomputed-statistic layers is left to future work.

3.2 Benchmark Substrate and Agent Architecture

The benchmark as a tool substrate. The benchmark’s tool layer—the same sixteen tools used in all experiments—is published as a Model Context Protocol (MCP) server [2], an open standard for connecting language-model agents to external tools; the agent under test is therefore one MCP *client* against a fixed tool server, task suite, and scorer, and any MCP-compatible agent can be evaluated against the identical tools. The server exposes only the MCP *tools* primitive—no resources or preloaded context—so the benchmark measures whether an agent can select the correct tool for a task from the tool descriptions alone. This scope is deliberate: the same substrate extends to the other MCP primitives (resources, prompts) and to richer agentic constructs such as subagent orchestration, which we leave to future evaluation work.

Figure 1 shows the system architecture. The agent operates in a ReAct-style loop [29]: it receives a query, reasons about tools, executes calls, observes results, and generates a response. The agent has access to sixteen tools: the twelve principal operations in Table 2, plus four auxiliary tools for layer discovery (`list_layers`, `get_legend`, `get_layer_bounds`) and erosion time-series statistics (`query_erosion_stats`).

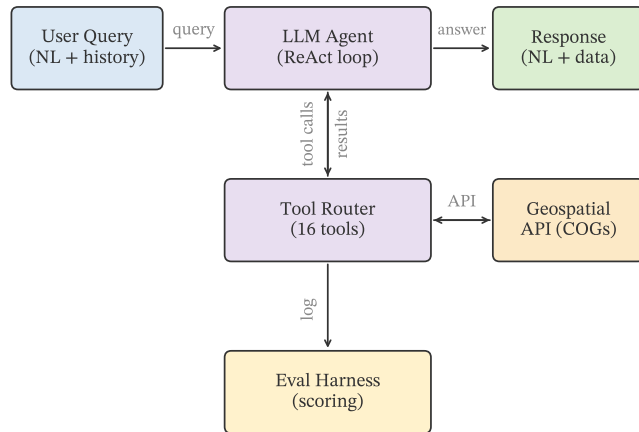


Figure 1: GeoNatureAgent Benchmark system architecture. The LLM agent interacts with a production API serving COG data; the eval harness logs all tool calls and scores responses. NL denotes natural language.

The design choice of structured tool calling over code generation is motivated by empirical evidence: GeoJSON Agents report 85–97% accuracy with function calling [18], while UnivEARTH reports only 33% when agents generate executable code [12].

Table 2: Principal agent tools. The first eight are domain-specific; the last four are adapted from GeoBenchX [14]. The agent additionally exposes four auxiliary tools not listed here (layer discovery and erosion statistics).

Tool	Purpose
<code>lookup_province</code>	Resolve province → boundary geometry
<code>lookup_municipality</code>	Resolve municipality (+ province hint)
<code>analyze_area</code>	Zonal statistics for indicator in AOI
<code>analyze_multi_layer</code>	Multi-indicator analysis in single call
<code>compare_areas</code>	Compare two areas on the same indicator
<code>find_top_n</code>	Rank provinces by indicator value
<code>toggle_layer</code>	Enable/disable map layer visibility
<code>generate_chart</code>	Bar/stacked-bar chart from data
<code>create_buffer</code>	Buffer geometry by distance (km)
<code>select_spatial_rel</code>	Select features by spatial predicate
<code>get_centroids</code>	Extract centroid coordinates
<code>reject_task</code>	Decline unsolvable tasks

Multi-backend inference client. Fair cross-model comparison requires normalizing heterogeneous API surfaces into a common interface. Our inference client routes models through two backends: a native Anthropic client for Claude and a LiteLLM-based client for Vertex AI Model-as-a-Service (MaaS) endpoints that expose an OpenAI-compatible API. The MaaS backend maps publisher-prefixed model identifiers (e.g., `meta/llama4-...`) to Vertex AI endpoints and normalizes response formats—including tool-call representations—into the shared schema consumed by the eval harness. Notably, Llama 4 models emit tool calls as Python-style function invocations (`<|python_start|>func(k=v)<|python_end|>`) rather than JSON; a dedicated AST-based parser extracts these into structured tool calls so that Llama 4 Scout’s responses are scored on the same footing as the JSON-native models. Each inference call uses a layered timeout architecture combining daemon-thread hard timeouts with exponential-backoff retry on rate-limit errors, enabling unattended execution of the full case-seed grid.

3.3 Evaluation Protocol

Each task is scored by eight check types (see Table 3 and Figure 2). A task *passes* only when **all** applicable checks pass (strict binary). We additionally compute partial-credit metrics for finer-grained analysis.

Continuous metrics. Beyond binary pass/fail, we report: *check score*—fraction of individual checks passed per case (partial credit); *keyword coverage*—fraction of `must_contain` terms found; *tool F1*—set-based F1 between expected and actual tools; and *cost utilisation*—ratio of actual to budgeted cost.

Error classification. When a task fails, the first failing check (in priority order: `tool_missing` → `chart_missing` → `wrong_data` → `rounds_exceeded` → `keyword_missing` → `forbidden_keyword`) determines the error category. This priority reflects diagnostic value: tool selection failures are the most informative for understanding

Table 3: Scoring checks. Each task specifies which checks apply; binary capability pass requires every applicable check except `max_cost_usd` to pass (Paragraph “Capability and cost as orthogonal axes”).

Check	Pass condition
<code>expected_tools</code>	Every expected tool was called (recall = 1.0). Extra tools do not cause failure. ^a
<code>expected_actions</code>	Every expected UI action was generated (recall = 1.0).
<code>must_contain</code>	Each required keyword appears in the answer (case-insensitive substring).
<code>must_not_contain</code>	Forbidden keyword does not appear in the answer.
<code>numeric_accuracy</code>	For each ground-truth entry, the label is found in the answer and the nearest percentage value is within tolerance. ^b
<code>chart_generated</code>	At least one chart URL was produced (only checked when <code>generate_chart</code> is an expected tool).
<code>max_rounds</code>	Agent loop iterations $\leq$ budget.
<code>max_cost_usd</code> ^c	Estimated token cost $\leq$ per-case budget. <i>Reported but not gated</i> on capability pass/fail.

^aTool F1 (precision, recall, F1 on unique tool sets) is computed as a continuous metric but is not used as a gate.

^bA 120-character window after the label is searched for the first float% pattern; tolerance defaults to ± 2 pp.

^cPer-case budgets default to \$0.10 in the v5 task set. The per-case cost is logged and used in cost-accuracy analysis but is excluded from the binary capability score; see “Capability and cost as orthogonal axes” below.

agent reasoning. Cases that would only fail `max_cost_usd` are not counted as failures for capability scoring; their cost is instead reported on the cost-accuracy axis (Section 4.4). This first-failing-check attribution necessarily imposes an ordering on error modes, so the resulting distribution (Table 8) should be read as the *dominant* failure mode per run rather than an exhaustive multi-label tally. The order-independent continuous metrics—check score, tool F1, and keyword coverage—do not depend on this ordering, and we report them alongside the binary pass precisely so that conclusions do not rest on the strict binary criterion alone.

Design rationale. Tool and action checks use *recall* as the gate rather than F1. An agent that calls all expected tools plus an exploratory extra tool should not be penalised—the key question is whether it performed the required operations. F1 is tracked as a separate metric for efficiency analysis.

Capability and cost as orthogonal axes. Per-case cost budgets are deployment thresholds, not measures of reasoning ability: a model that pays more in tokens has not necessarily reasoned less correctly. Conflating the two into a single binary score forces the reader to interpret a model’s accuracy through the lens of one specific, unjustified price threshold. We therefore decouple them: the binary leaderboard scores capability only (does the agent solve the case, evidenced by tool selection, keywords, and numeric accuracy), and per-case cost is reported alongside as a Pareto axis (Section 4.4).

This separation follows HELM [4], GAIA [20], and other modern agent benchmarks that report cost and latency as orthogonal dimensions rather than gates. Aggregated cost remains visible in the leaderboard table; a reader who cares about a specific per-case budget can recover the cost-gated number from the supplementary per-case CSV (column `passed_with_cost_gate`).

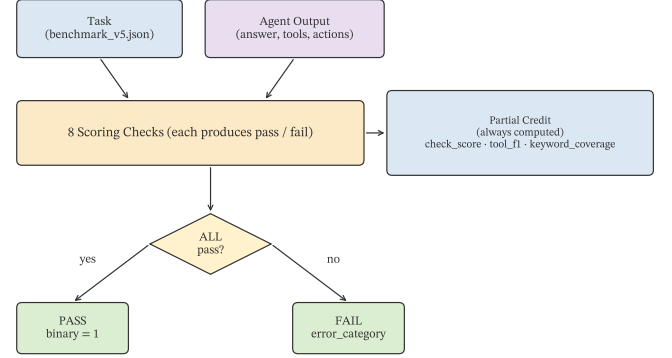


Figure 2: Scoring pipeline. Each case is evaluated by up to eight checks; binary pass requires all to pass. Partial-credit metrics (check score, tool F1, keyword coverage) are always computed.

4 Experiments and Results

4.1 Experimental Setup

We evaluate nine LLMs on GeoNatureAgent Benchmark v5 (93 tasks, 18 categories) across Vertex AI Model-as-a-Service, OpenRouter, and the Anthropic Messages API (see Table 4). All experiments use the v3 system prompt (a zero-shot ReAct-style instruction; the full prompt is committed to the repository at `api/agent/prompts/v3.md`), `max_turns=10`, `max_tokens=4096`, and `temperature=1.0`. We use temperature-1.0 to expose run-to-run variance and stress-test tool-use reliability under realistic sampling noise rather than the deterministic temperature-0 regime that overstates production stability. The prompt is held constant across all models to isolate capability from prompt-engineering effort. Chain-of-thought, few-shot, and self-reflection variants are distinct research questions and are left for future work. Experiments are run sequentially inside a single Cloud Run Job to ensure identical infrastructure conditions.

4.2 Aggregation Protocol

To control for run-to-run variance under temperature-1.0 sampling, every model is evaluated on the full v5 case set (93 tasks) under three random seeds chosen without any criteria, {42, 1337, 2024}, where the underlying inference API exposes a `seed` parameter (Vertex and OpenRouter endpoints). The Anthropic Messages API does not implement deterministic seeding, so Claude Sonnet 4 is evaluated under two independent temperature-1.0 samples drawn under the same configuration; this yields a valid run-to-run variance estimate but is not bit-for-bit replayable. Reported accuracies are the per-seed mean with one standard deviation, and per-category accuracies are the unweighted mean across the available seeds. Each Cloud Run output directory contains a `results.jsonl` where

Table 4: Models evaluated. Six via Vertex AI (five MaaS plus Gemini native), two via OpenRouter, Claude via the Anthropic Messages API.

Model	Provider	Params	Access
DeepSeek V3.2	DeepSeek AI	671B MoE	Vertex MaaS
GLM-5	Zhipu AI	—	Vertex MaaS
Gemini 2.5 Pro	Google	—	Vertex native
Claude Sonnet 4	Anthropic	—	Anthropic API
GPT-4o	OpenAI	—	OpenRouter
Qwen3-235B	Alibaba	235B MoE	Vertex MaaS
GPT-OSS-120B	OpenAI	120B	Vertex MaaS
Llama 4 Scout	Meta	109B MoE	Vertex MaaS
Gemma-3-27B	Google	27B	OpenRouter

Vertex MaaS models run on provider-managed serving stacks whose quantization and datacenter region are neither disclosed nor controlled by us; absolute cross-provider latency and dollar cost therefore carry this confound. The tool interface is identical across all models, so capability differences reflect the model rather than the tool layer, but serving-stack differences are not eliminated and should be borne in mind when reading absolute cost figures.

every row carries its own seed, case_id, and experiment_id, so any leaderboard cell can be traced back to its Cloud Run output directory.

The mapping from each model’s leaderboard entry to its specific Cloud Run output directory is committed in paper/final_results/sources.yaml. A single deterministic script (scripts/compile_final_results.py) reads this manifest, pulls each results.jsonl from GCS, and emits the per-model leaderboard, per-category breakdown, and per-case matrix as CSV artifacts that are themselves committed to the repository. These CSVs are the single source of truth for every number reported in this paper; a reviewer can regenerate them from the public bucket with some commands and diff-check against the committed copies.

4.3 Overall Results

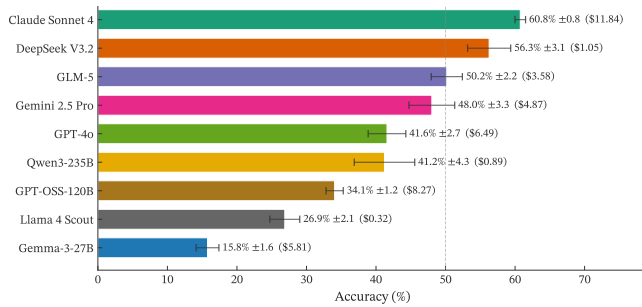
**Figure 3: GeoNatureAgent Benchmark v5 leaderboard. Accuracy (%) with total cost in parentheses. Dashed line at 50%.**

Table 5 and Figure 3 summarize the main results aggregated across seeds. **Claude Sonnet 4** leads on capability at $60.8\% \pm 0.8\%$, with **DeepSeek V3.2** a close second at $56.3\% \pm 3.1\%$. **GLM-5** ($50.2\% \pm 2.2\%$) and **Gemini 2.5 Pro** ($48.0\% \pm 3.3\%$) form the second tier. **GPT-4o** ($41.6\% \pm 2.7\%$) and **Qwen3-235B** ($41.2\% \pm 4.3\%$) are effectively

Table 5: Capability leaderboard: nine models on 93 GeoNatureAgent Benchmark v5 tasks. Accuracy is the per-seed mean with one standard deviation, decoupled from cost (Section 3.3). Cost is the mean total cost of one seed run (93 cases); \$/case is the per-call mean.

#	Model	Accuracy	Cost	\$/case
1	Claude Sonnet 4 [†]	$60.8\% \pm 0.8\%$	\$11.84	0.127
2	DeepSeek V3.2	$56.3\% \pm 3.1\%$	\$1.05	0.011
3	GLM-5	$50.2\% \pm 2.2\%$	\$3.58	0.038
4	Gemini 2.5 Pro	$48.0\% \pm 3.3\%$	\$4.87	0.052
5	GPT-4o	$41.6\% \pm 2.7\%$	\$6.49	0.070
6	Qwen3-235B	$41.2\% \pm 4.3\%$	\$0.89	0.010
7	GPT-OSS-120B	$34.1\% \pm 1.2\%$	\$8.27	0.089
8	Llama 4 Scout	$26.9\% \pm 2.1\%$	\$0.32	0.003
9	Gemma-3-27B	$15.8\% \pm 1.6\%$	\$5.81	0.062

[†]Claude Sonnet 4 is reported from two temperature-1.0 samples (the Anthropic Messages API does not implement deterministic seeding); the other eight models are evaluated under three seeds. The variance estimate for Claude is therefore based on fewer observations and should be interpreted as an indicative measure rather than a precise one. We do not draw a third Claude sample because its two samples already exhibit the tightest run-to-run spread in the group (± 0.8 pp) while incurring the highest per-run cost (\$11.84); an additional sample would add disproportionate expense for limited variance information.

tied in the third tier, followed by **GPT-OSS-120B** ($34.1\% \pm 1.2\%$) and **Llama 4 Scout** ($26.9\% \pm 2.1\%$). **Gemma-3-27B** floors the leaderboard at $15.8\% \pm 1.6\%$, partly a harness artifact: it emits tool calls as “tool_code markdown blocks rather than the parsed structured format, and extending the parser to accept that dialect (as with Claude Sonnet 4.5, Section 4.8) roughly tripled its accuracy from a raw 5.7% and dropped its no-tool-call rate from 96% to 68%. It remains last on residual unparsed blocks and over-refusal—a weak but now fairly-measured model. Only two models exceed 55% accuracy, and none reach 65%, confirming that environmental geospatial tool orchestration remains a significant challenge even for the strongest models. Inter-seed standard deviations are tight (0.8–4.3 percentage points), so the inter-model ordering is robust to sampling noise.

Three findings stand out. First, **the top of the leaderboard is bimodal but close**: Claude Sonnet 4 and DeepSeek V3.2 are separated by only 4.5 percentage points (Welch’s $t = 2.40$, $df \approx 2$, $p \approx 0.13$)—within statistical noise at our sample size—while both lead the third-place GLM-5 by more than 6 pp ($t \geq 2.76$). The capability gap between the top two and the rest of the models is robust: Claude Sonnet 4 vs. Llama 4 Scout yields $t = 25.0$, and every adjacent pair below the top two is significant at $p < 0.1$ except GLM-5 vs. Gemini 2.5 Pro, which are statistically indistinguishable. Second, **accuracy tracks scale strongly within our small open-weight sample and across all three seeds**: Llama 4 Scout (109B MoE active parameters) trails Qwen3-235B, which trails DeepSeek V3.2 (671B MoE); the limitation to this claim is the number of models tested (quantified below). Third, **closed-source frontier models do not categorically dominate open-weight peers**: Claude leads on capability but only marginally above DeepSeek; Gemini 2.5 Pro sits in third position, below both open-weight contenders. As we show in Section 4.4, this pattern is reinforced on the cost axis, where

open-weight models occupy most of the cost-accuracy Pareto frontier.

Quantifying the scale trend. Restricting to the five open-weight models with disclosed parameter counts (Gemma-3-27B, 27B; Llama 4 Scout 109B; GPT-OSS-120B, 120B; Qwen3-235B, 235B; DeepSeek V3.2, 671B), accuracy rises monotonically with total parameters: Spearman $\rho = 1.0$ and Pearson $r = 0.98$ (on log total parameters), with the ranking identical on each of the three seeds individually (verified by scripts/scaling_stats.py). With Gemma-3-27B added the sample reaches $n = 5$, so the exact two-sided permutation $p = 2/5! = 0.017$ now clears the conventional 0.05 threshold—the smallest attainable p at $n = 4$ was 0.083. The limitation to this claim remains the number of models tested and that all are distinct in architecture and training, so we read it as a strong within-sample trend rather than a general scaling law.

Partial credit tells a different story. Even failing cases often demonstrate partial competence: correct tool selection but missing a keyword, or correct keywords but an extra unnecessary tool call. This gap between binary accuracy and partial-credit scores (see Figure 4) suggests that many failures are “near-misses” addressable through prompt engineering.

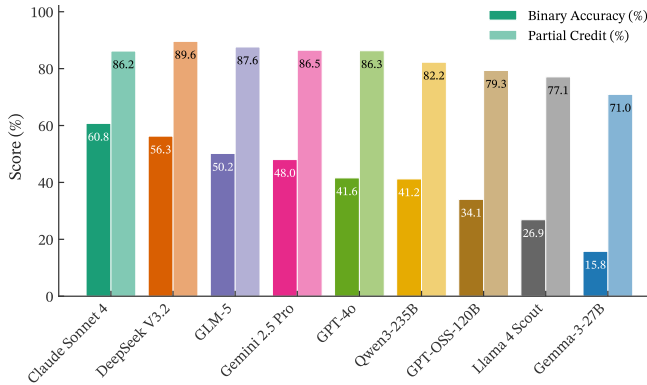


Figure 4: Binary accuracy vs partial-credit check score. The gap indicates many “near-miss” failures.

Bootstrap confidence intervals. To quantify how much of the inter-model spread is small-sample noise, we compute 95% bootstrap confidence intervals on each model’s accuracy by resampling the 93 cases (2,000 resamples; Figure 5). The intervals are wide and overlap substantially between adjacent models—Claude Sonnet 4 [51.1, 69.9] and DeepSeek V3.2 [47.7, 64.5] overlap almost entirely, as do GLM-5 [41.2, 58.1] and Gemini 2.5 Pro [39.8, 56.3]—so adjacent-model and category-level rank claims should be read as indicative rather than definitive at $n = 93$. Only the aggregate top-versus-bottom separation (Claude Sonnet 4 vs. Llama 4 Scout [19.7, 33.7]) sits well outside any overlap.

4.4 Cost-Accuracy Analysis

Cost per case varies by more than an order of magnitude across the different models, from \$0.003 for Llama 4 Scout to \$0.127 for Claude Sonnet 4 (see Table 5). The cost-accuracy Pareto frontier

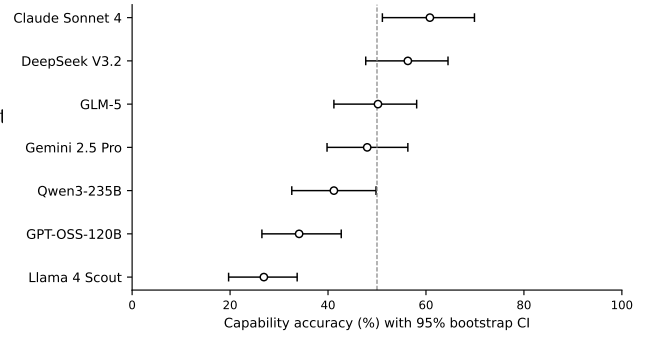


Figure 5: Capability accuracy with 95% bootstrap confidence intervals (3 seeds, 2,000 resamples); seven of the nine models are shown (GPT-4o and Gemma-3-27B omitted for readability). Overlapping intervals show that adjacent-model and category-level claims rest on small samples.

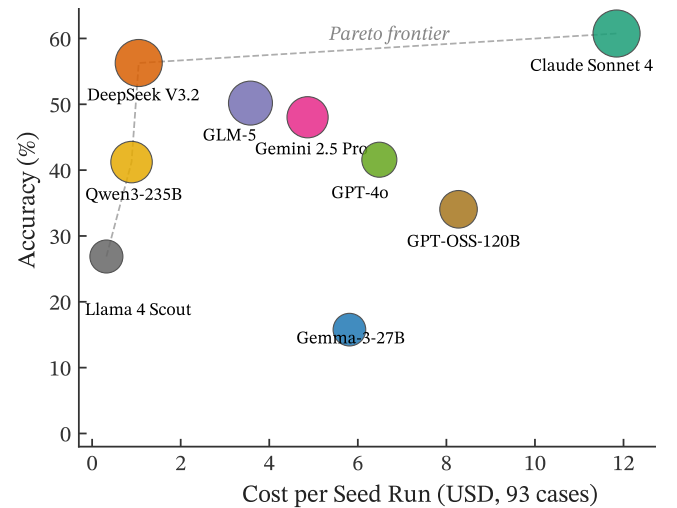


Figure 6: Cost-accuracy trade-off. Bubble size is proportional to total tokens; cost is the mean per-seed run cost (93 cases). The Pareto frontier (dashed line) runs Scout → Qwen3-235B → DeepSeek V3.2 → Claude Sonnet 4. Three of the four models on this frontier are open-weight.

(Figure 6) comprises four models: **Llama 4 Scout** (cheapest, 26.9% capability), **Qwen3-235B** (41.2% at \$0.010/case), **DeepSeek V3.2** (56.3% at \$0.011/case), and **Claude Sonnet 4** (highest capability at 60.8%, \$0.127/case). Three of these four Pareto-frontier models are open-weight; the remaining five models are all dominated (every Pareto-frontier point matches or exceeds their accuracy at lower cost): closed-source Gemini 2.5 Pro and GPT-4o, and open-weight GLM-5, GPT-OSS-120B, and Gemma-3-27B. The DeepSeek V3.2 entry is particularly notable: it delivers 93% of Claude’s capability (56.3% vs. 60.8%) at 11.6× lower per-case cost, making it the strongest single-model recommendation for cost-sensitive deployments. The dominated interior includes both mid-tier options priced between DeepSeek and Claude (GLM-5, Gemini 2.5 Pro, GPT-4o,

GPT-OSS-120B) and Gemma-3-27B, whose low accuracy (15.8%) leaves it dominated by the far cheaper Llama 4 Scout.

Table 6: Projected per-query inference cost at 1,000 queries/day. Daily and monthly data extrapolate the per-case cost reported in Table 5. Sorted by capability accuracy.

Model	Accuracy	Daily	Monthly
Claude Sonnet 4	60.8%	\$127	\$3,810
DeepSeek V3.2	56.3%	\$11	\$330
GLM-5	50.2%	\$38	\$1,140
Gemini 2.5 Pro	48.0%	\$52	\$1,560
GPT-4o	41.6%	\$70	\$2,100
Qwen3-235B	41.2%	\$10	\$300
GPT-OSS-120B	34.1%	\$89	\$2,670
Llama 4 Scout	26.9%	\$3	\$90
Gemma-3-27B	15.8%	\$62	\$1,860

Table 6 projects per-query costs to a production deployment serving 1,000 queries/day. The capability leader (Claude Sonnet 4, 60.8%) costs \$127/day, while the second-place DeepSeek V3.2 (56.3%) costs \$11/day—a 4.5-percentage-point capability gap for an 11.6× price difference. For most production deployments, the right operating point sits on the open-weight Pareto frontier; the rare scenarios that justify the premium price are those where the absolute capability ceiling matters more than throughput economics. These dynamics align with prior cost-efficient findings such as Geo-OLM [25], which shows that small specialized models can match frontier model performance at a fraction of frontier cost; here, the same pattern emerges among general-purpose models at the larger scale required for environmental analysis.

From cost to carbon. Extending the cost analysis to environmental footprint, we estimate energy and carbon per query from token usage (blended 0.40 Wh per 1,000 tokens; global-average grid intensity 400 gCO₂/kWh; an order-of-magnitude estimate, not a measurement). Footprint broadly rises with capability—the most capable model (Claude Sonnet 4) emits an estimated 6.4 gCO₂ per case versus 3.2 gCO₂ for the least capable (Llama 4 Scout), a 2× range (Table 7)—but it is driven by *tokens*, not accuracy directly: DeepSeek V3.2 is token-heavy and so nearly matches Claude’s per-case carbon (6.4 gCO₂) despite its 11.6× lower dollar cost, decoupling the carbon and dollar axes. For an *environmental* benchmark this makes footprint a first-class deployment axis alongside accuracy and cost.

Figure 7 further illustrates this relationship, showing that token consumption does not predict accuracy—models that generate more tokens do not necessarily achieve higher scores.

4.5 Failure Analysis

4.5.1 Universally Hard Tasks. Several categories sit in the 0–25% range (see Figure 8), revealing systematic limitations:

Close-value comparison (case V5_21): two near-identical values (Murcia 68.5% vs. Córdoba 68.4%) lead models to report a directional difference, while the directional sub-case (V5_22) is partly solvable; as a single item we treat this as indicative only, and the v6

Table 7: Estimated energy and carbon per case (tokens × 0.40 Wh/1k; 400 gCO₂/kWh). Order-of-magnitude estimate; footprint is token-driven.

Model	Accuracy	gCO ₂ /case
Claude Sonnet 4	60.8%	6.44
DeepSeek V3.2	56.3%	6.42
GLM-5	50.2%	5.61
Gemini 2.5 Pro	48.0%	4.92
GPT-4o	41.6%	3.56
Qwen3-235B	41.2%	4.99
GPT-OSS-120B	34.1%	4.09
Llama 4 Scout	26.9%	3.16
Gemma-3-27B	15.8%	3.09

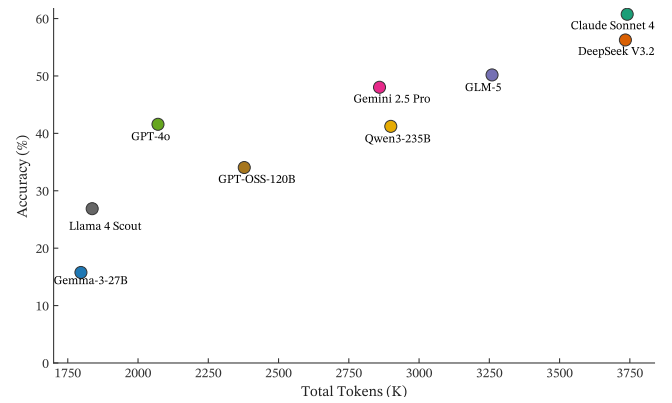


Figure 7: Token consumption vs. accuracy. Token volume is a poor predictor of performance.

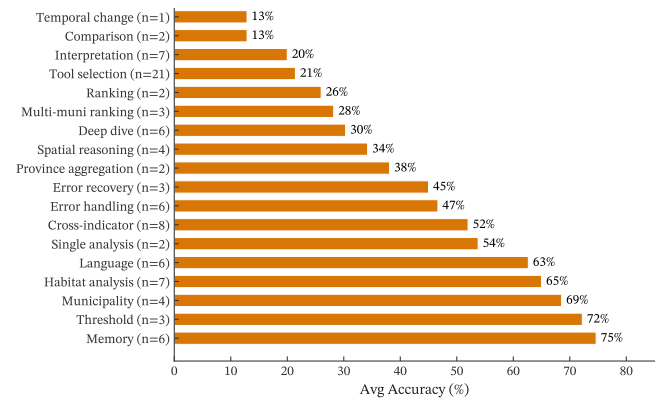


Figure 8: Category difficulty —mean accuracy across all nine models, sorted by difficulty.

expansion (Section 4.6) probes it across ten comparison tasks spanning value gaps from 0.6 to 45 percentage points. **Ranking** (0–67% across models): Bottom-N and Top-N province queries divide the models into two groups—GLM-5 (67%) and Claude (50%) handle them while Gemini, GPT-OSS-120B, and Llama 4 Scout floor at 0%,

with errors split between calling the wrong tool (`analyze_area` instead of `find_top_n`) and confabulating rankings from training data. **Single analysis** (17–100%): DeepSeek V3.2 solves it perfectly (100%) while Gemini 2.5 Pro floors at 17%. **Province aggregation** (0–75%): wide variance, with Claude (75%) and Gemini 2.5 Pro (67%) highest but Qwen3-235B (0%) and DeepSeek (17%) much lower (see Figure 10).

Conversely, the easiest categories across the nine models are **memory/multi-turn** (mean 75%), **threshold** (72%), and **municipality** (69%), with **habitat analysis** (mean 65%; Claude 100%, GLM-5 81%) and **language** (mean 63%, with Llama 4 Scout, DeepSeek, and Claude all at 75–78%) close behind. Models broadly sustain multi-turn context and parse Galician/Basque inputs but cannot reliably perform the multi-step numeric reasoning that ranking and comparison demand.

4.5.2 Model-Specific Patterns. **DeepSeek V3.2** (56.3%): leader, with the most balanced profile—above 70% on cross-indicator synthesis, memory, multi-municipality ranking, single analysis, spatial reasoning, threshold, and municipality categories. Notably the only top-three model to break 30% on the hard comparison category (33%; among the rest only GPT-OSS-120B, at 50%, also exceeds zero). **GLM-5** (50.2%): strong on cross-indicator (88%), habitat (81%), memory (83%), municipality (92%), and threshold (78%); weaker on tool selection (27%) and a 0% floor on comparison (though it tops the cohort on ranking at 67%). **Gemini 2.5 Pro** (48.0%): the highest-accuracy closed-source model in the group, with habitat (86%), memory (78%), and error handling (72%) among its strengths; notably weak on ranking and temporal change (both 0%). **Claude Sonnet 4** (60.8%): the capability leader, with the most uniformly strong per-category profile in the group. Claude reaches 100% on habitat analysis, memory/multi-turn, and municipality; 94% on cross-indicator synthesis (the highest in the evaluation group); 83% on threshold; and 75% on language, spatial reasoning, and province aggregation. Its principal weakness is the hard comparison category (0%, shared with four other models; only DeepSeek and GPT-OSS-120B exceed zero) and tool selection (33%, tied with DeepSeek but well below its other categories). Per-token cost is high (\$0.127/case), placing Claude on the upper-right of the Pareto frontier: the most capable single point, but at 11.6× the price of the next-best Pareto point (DeepSeek). **Qwen3-235B** (41.2%): strongest on threshold (100%), habitat (76%), memory (83%), and deep dive (56%); weakest on tool selection (17%) and province aggregation (0%). **GPT-OSS-120B** (34.1%): moderate across categories with no standout strengths; threshold (89%) is its only category above 70%. **Llama 4 Scout** (26.9%): trails the group, with successes concentrated in language (78%), memory (72%), and error recovery (67%) but a near-floor on tool selection (6%) and cross-indicator synthesis (29%) (see Figure 10).

4.5.3 Failure-mode distribution. Across all 1,429 failed model–seed–case runs, failures concentrate in tool selection rather than data handling. Tabulating each failed run by its scorer-assigned error category gives: tool missing (an expected tool was never called) 68.0%; keyword missing (the answer omits a required term) 19.3%; wrong data (an incorrect value) 9.7%; rounds exceeded 1.3%; forbidden keyword 1.3%; and chart missing 0.3% (Table 8). That over two-thirds

of all failures are tool-missing—not wrong-data—reframes the difficulty: models fail predominantly by not invoking the right tool, not by mis-reading correct tool output, consistent with tool selection being the most discriminative category (Section 4.7). At the case level, eight cases are unsolved by *every* model in *every* seed (V5_21, V5_23, V5_25, V5_61, V5_70, V5_71, V5_76, V5_77), spanning comparison, deep-dive, tool-selection, and interpretation—a more precise statement than a single “unsolved” task, though V5_21’s close-value difficulty is re-attributed to tool composition rather than numeric discrimination in Section 4.6.

Table 8: Failure-mode distribution over all 1,429 failed runs (nine models).

Failure mode	% of failures
Tool missing (expected tool not called)	68.0
Keyword missing (answer lacks required term)	19.3
Wrong data (incorrect value)	9.7
Rounds exceeded	1.3
Forbidden keyword	1.3
Chart missing	0.3

4.6 v6 Expansion: Close-Value Comparison Revisited

The close-value finding above rests on a single case (V5_21), so we ran a ten-task expansion set (v6) of comparison-heavy queries over the Portuguese BigEarthNet land-cover layer, spanning ground-truth value gaps from 0.6 to 45.1 percentage points. It includes a deliberate large-gap CONTROL (V6_05, 45.1 pp, expected to be trivial) and three tasks that require composing a per-district sum before comparing. All nine leaderboard models were run under the identical three-seed, temperature-1.0 protocol (aggregated by `scripts/aggregate_v6.py`; Table 9).

Three findings refine the v5 result. First, **the “unsolved by every model” statement does not generalize**: under strict (raw) scoring Gemini 2.5 Pro solves the majority of comparison runs (67%, including the control at 100%) and GPT-4o reaches 50%, so at least two models reliably discriminate near-equal values when the task is well grounded. Second, **difficulty does not track the value gap** (Figure 9, left): pooled raw accuracy is essentially flat across gaps (26% at 0.6 pp, 37% at 1.2 pp, 19% at 2.1 pp), and the large-gap control is *no easier* (22% raw) than the close cases, while the two sum-then-compare tasks are the hardest of all (0% raw across every model and seed). The deficit is therefore not near-equal numeric discrimination but the multi-step tool composition the Portugal layer demands: served only by `analyze_area`, each comparison requires several sequential calls, and models frequently exhaust their turn budget. Third, **this manifests as a raw-versus-capability gap** (Figure 9, right): capability scoring—which excuses budget/round exhaustion exactly as in the v5 leaderboard—lifts comparison accuracy to 67–83% for the top models, so most of their comparison “failures” are timeouts mid-composition rather than wrong answers. Claude Sonnet 4 is the extreme case: 83% comparison capability but 0/18 raw, its passes being entirely round-exhaustion excusals.

The expansion thus supports treating the v5 close-value case as indicative rather than a fundamental limitation, and re-attributes the residual difficulty to tool composition and turn budget—consistent with tool-missing and round exhaustion dominating the corpus-wide failure mix (Table 8).

Combined 103-task metric. Beyond the per-suite view, folding the expansion into the main suite yields an inclusive 103-task capability leaderboard (last column of Table 9; computed by `scripts/aggregate_combined.py`): Claude Sonnet 4 61.7%±0.7%, DeepSeek V3.2 57.9% ± 3.9%, GLM-5 52.4% ± 2.6%, down to Gemma-3-27B 14.6% ± 1.0%. The v5 ordering is preserved throughout (only Qwen3-235B and GPT-4o swap, within noise), so including the new tasks confirms rather than alters the headline findings. Per-suite v6 accuracies remain reported above because the expansion is deliberately comparison-heavy and not distribution-matched to the main suite; the 93-task leaderboard (Table 5) remains the reference point comparable with prior versions of this work.

Table 9: v6 expansion (10 comparison-heavy tasks, 3 seeds) and the combined 103-task suite. Cap. is capability scoring (budget/round gates excused, as in the v5 leaderboard); raw is strict pass. 103-task comb. is capability accuracy over the full v5+v6 suite (per-seed mean), the inclusive headline metric for this paper.

Model	Cap.	Raw	Comp. cap/raw	103-task comb.
Qwen3-235B	73%	37%	78/22%	44.3%
GLM-5	73%	37%	83/33%	52.4%
DeepSeek V3.2	73%	27%	83/17%	57.9%
Claude Sonnet 4	73%	20%	83/0%	61.7%
GPT-4o	63%	50%	72/50%	43.7%
Gemini 2.5 Pro	60%	60%	67/67%	49.2%
GPT-OSS-120B	20%	20%	11/11%	32.7%
Llama 4 Scout	17%	13%	11/6%	25.9%
Gemma-3-27B	3%	3%	0/0%	14.6%

4.7 Category-Level Analysis

Figure 10 shows mean accuracy by category across all nine models. Two structural findings emerge:

(1) Tool selection variance is the dominant cross-model spread. The 21-case tool selection category is the largest and the most discriminative: Claude (33%), DeepSeek (30%), GLM-5 (27%), GPT-OSS-120B (25%) cluster within an 8-point band, while Qwen3-235B (17%) and Llama 4 Scout (6%) trail. **(2) Model strengths are domain-specialized rather than uniform.** No model dominates every category. Claude leads on habitat analysis (100%), memory (100%), municipality (100%), and cross-indicator synthesis (94%). DeepSeek leads on single analysis (100%), spatial reasoning (83%), and multi-municipality ranking (78%). GLM-5 tops the cohort on ranking (67%) and is the only mid-pack model to clear 65% on cross-indicator (88%). Qwen3-235B leads on threshold (100%), Gemini on error handling (72%), and GPT-OSS-120B on the comparison category (50%, via the directional sub-case). A practitioner optimizing deployment for a specific category should consult the per-category breakdown rather than the headline accuracy alone—no single

model is uniformly the best, and the right choice depends on which task patterns dominate the target workload.

4.8 Prompt-version sensitivity and reproducibility

Two controls govern how any tool-calling leaderboard should be read: how much the system prompt moves scores, and whether the harness reproduces. We probe both by re-running the v5 suite under the paper’s v3 prompt on three current models (single seed; Table 10). **Prompt sensitivity:** scores depend strongly on prompt version—moving from the runner’s default (v2) to the paper’s v3 prompt raises Claude Haiku 4.5 by +22.6 pp (25.8% → 48.4%) and Gemini 2.5 Pro by +11.8 pp (30.1% → 41.9%), a swing larger than most model-to-model gaps in Table 5. We therefore fix the v3 prompt across all models for comparability and recommend future runs do the same. **Reproducibility:** the fresh Gemini 2.5 Pro run scores 41.9%, inside the paper’s three-seed 95% bootstrap interval ([39.8, 56.3], mean 48.0%), confirming the harness reproduces to within sampling noise. **New models** enter for coverage: Claude Haiku 4.5 reaches 48.4%, matching the mid tier at a fraction of the flagship cost, while Claude Sonnet 4.5 scores only 30.1% on binary pass despite an 85% partial-credit score—a tool-call/output-format mismatch under our scorer rather than reduced capability (flagged for verification; single seed). These rows are reported alongside, not as a re-ranking of, the accepted nine-model leaderboard (Table 5).

Table 10: Prompt-version sensitivity and reproducibility (v3 prompt, single seed). The v2→v3 shift exceeds most model-to-model gaps in Table 5, motivating a fixed prompt; the Gemini 2.5 Pro re-run reproduces the paper within its CI. New-model rows are reported alongside—not a re-ranking of—the accepted leaderboard.

Model	Acc. (v3)	Note
Claude Haiku 4.5	48.4%	new; +22.6 pp vs v2; matches mid tier
Gemini 2.5 Pro (re-run)	41.9%	+11.8 pp vs v2; reproduces paper (CI [39.8, 56.3])
Claude Sonnet 4.5	30.1%	85% partial-credit; format mismatch (to verify)

5 Discussion

5.1 Comparison with Prior Benchmarks

GeoNatureAgent Benchmark’s 60.8% best-model accuracy is notably lower than GeoBenchX and GeoJSON Agents (85–97%). Table 11 summarizes the key differences.

Four factors explain the accuracy gap: (1) real API interaction vs. local file operations; (2) environmental domain complexity (geographic knowledge, data limitations, display-only layers); (3) multi-turn conversation history; and (4) strict binary evaluation. Our results align with harder benchmarks: MapEval (< 67%) [9], SpatialLab (55%) [28], and UnivEARTH (33%) [12].

The GeoBenchX comparison is particularly instructive because both benchmarks share the structured tool-calling paradigm. The roughly 30-percentage-point accuracy gap between GeoBenchX

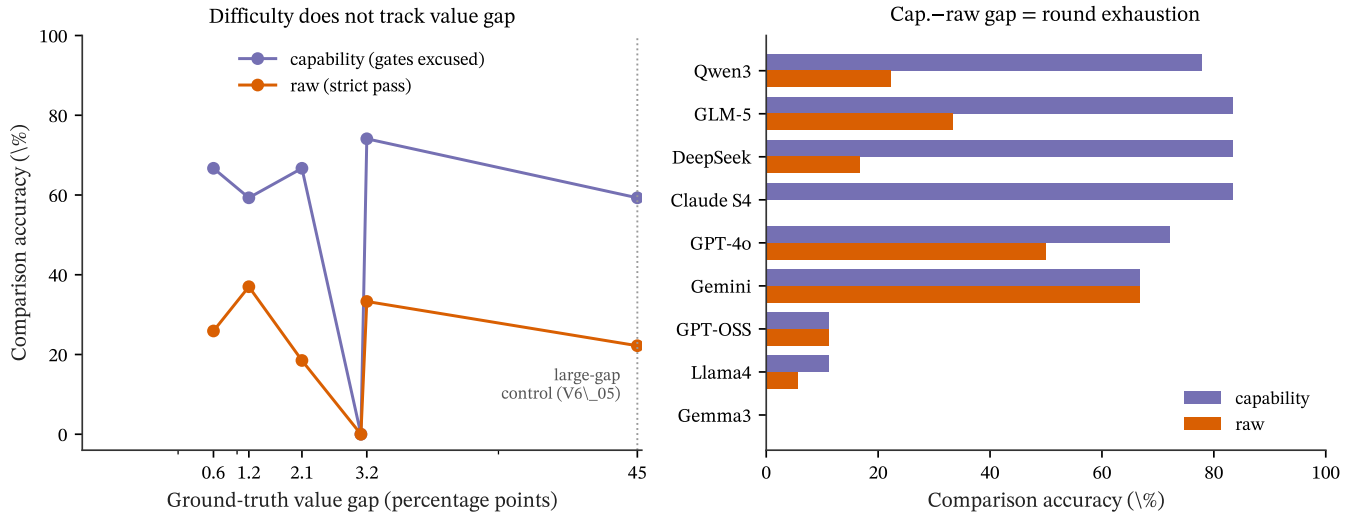


Figure 9: v6 comparison tasks. *Left:* pooled comparison accuracy vs. ground-truth value gap under raw and capability scoring—accuracy is flat in the gap and the 45 pp control is no easier than the close cases, so the difficulty is not near-equal discrimination. *Right:* per-model comparison accuracy, capability vs. raw; the gap between the bars is round/budget exhaustion while composing repeated `analyze_area` calls.

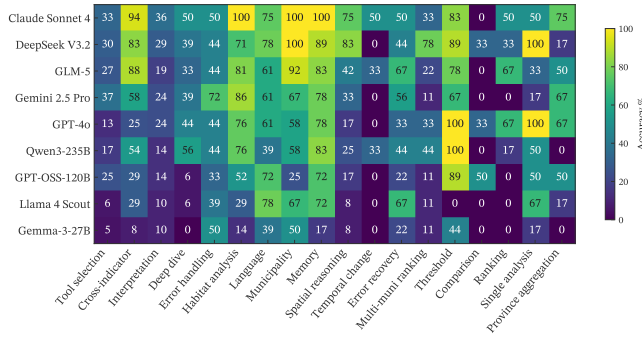


Figure 10: Accuracy by category across the nine models.

and our benchmark is consistent with general-purpose geospatial benchmarks overestimating agent capability for domain-specific workflows. We caution, however, that this comparison does not control for differences in task sampling or scoring rubric: part of the gap may reflect our stricter binary criterion rather than intrinsic task difficulty, so we report it as indicative rather than conclusive. To mitigate this, future versions of the benchmark will add tolerance- and range-based scoring—crediting answers that fall within an accepted interval of the ground truth—so that near-equal comparisons and numeric estimates are measured for correctness rather than penalized by a strict binary criterion. Environmental tasks require geographic knowledge (e.g., which provinces have erosion data), awareness of data limitations (display-only layers that cannot be analyzed), multi-step reasoning across indicators, and temporal awareness—challenges absent from generic filter-merge-visualize pipelines. This motivates domain-specific benchmarks as a necessary complement to general-purpose evaluations.

Table 11: Comparison of structured tool-calling geospatial benchmarks.

	GeoBenchX	GeoNatureAgent Benchmark
Tasks	202	93
Categories	5	18
Tools	24 primitives	12 domain ops
Tool abstraction	Low-level	High-level
Data source	Local files	Cloud API (COG + JSON)
Domain	General GIS	Environmental
Countries	—	Spain + Portugal
Indicators	—	3
Multi-turn	No	Yes
Best accuracy	85–90%	60.8%
Evaluation	LLM-as-judge	Metric-based

5.2 Structured Tool Calling vs. Code Generation

Our 60.8% best-model accuracy on real API tool calls compares favorably with UnivEARTH’s 33% on code generation [12]. While not directly comparable, the structured approach constrains the output space, eliminating the 58% code execution failure rate that UnivEARTH reports. This provides further evidence that production environmental systems should prefer structured APIs over free-form code generation.

5.3 Cross-Dataset Evaluation: BigEarthNet V2

GeoNatureAgent Benchmark is designed to be domain- and geography-agnostic. To demonstrate extensibility beyond Spanish CO₂ and erosion indicators, we incorporate BigEarthNet V2 [6] as a third data domain covering a different country. BigEarthNet V2

provides 549k labeled Sentinel-2 patches with CLC2018 19-class labels; we aggregate the 75k+ Portuguese patches across 9 mainland districts into a 7-class land cover distribution (Coniferous Forest, Broadleaf Forest, Shrubland, Grassland, Sparse Vegetation, Water, and Urban) queryable via the same `analyze_area` tool used for Spanish indicators.

This integration tests whether agents generalize from Spanish environmental data to a different country and data domain. The 20 BigEarthNet benchmark cases (Table 12) span seven task patterns: single-district breakdown, cross-indicator synthesis (Portuguese land cover with Spanish CO₂/erosion), multi-turn recall, chart generation, legend and layer discovery, temporal context, and two-district comparison. Unlike the CO₂ and erosion indicators, where ground truth depends on live raster geometry, BigEarthNet cases have fixed numeric ground truth derived from pre-computed district-level statistics, enabling deterministic scoring.

Table 12: BigEarthNet V2 benchmark cases by task pattern. All cases use Portuguese districts with pre-computed ground truth.

Task Pattern	Cases
Single-district land cover breakdown	3
Cross-indicator (PT land cover + ES)	3
Multi-turn recall from session history	2
Chart generation	2
Legend / list layers discovery	2
Deep-dive multi-indicator profile	4
Two-district comparison	4

While Table 12 lists the 20 purpose-built BigEarthNet cases, Portuguese land cover data also appears in 4 additional cases across other categories (tool selection, interpretation, and error recovery), yielding 24 BigEarthNet-related cases in total. Aggregated across seeds, **Claude Sonnet 4** achieves the highest mean accuracy on this subset (100% on the dedicated habitat-analysis cases and 94% on cross-indicator cases that combine Portuguese land cover with Spanish indicators), followed by **GLM-5** (habitat 81%, cross-indicator 88%) and **DeepSeek V3.2** (habitat 71%, cross-indicator 83%). Claude’s standout performance on the BigEarthNet cases—consistent with its top position on the overall capability leaderboard—suggests that its instruction-following and JSON-handling reliability are particularly well-suited to the pre-computed district-level indicator workflow. The BigEarthNet subset also illustrates a broader point: adding a third indicator and a second country required no engine changes, validating the framework’s domain- and geography-agnostic claim.

The inclusion of BigEarthNet V2 as a queryable layer rather than merely a validation dataset transforms the benchmark’s geographic scope from Spain-only to cross-country (Spain + Portugal), and its indicator count from two to three. This validates the claim that the framework generalizes to new data sources: adding a new indicator required only a JSON statistics file, a prompt update, and new benchmark cases—no changes to the scoring engine or evaluation protocol.

5.4 Limitations

Small samples: with 93 tasks across 18 categories, roughly half the categories contain three or fewer instances, so category-level and failure-mode claims are indicative rather than definitive (Section 4); the aggregate leaderboard is better powered than any single category. **Category imbalance:** counts are deliberately uneven (21 tool-selection cases vs. 1–2 for niche categories), reflecting capability importance but limiting per-category statistical power. **Serving heterogeneity:** models are served across heterogeneous, provider-managed stacks whose quantization and region are not controlled (Table 4), confounding absolute latency and cost though not the shared tool interface. **Single-region data:** indicators cover only Spain and Portugal, so geographic generalization is untested. **Evaluation strictness:** binary pass/fail may understate capability (Section 3.3 describes the partial-credit metrics that address this), and our strict rubric may inflate the measured gap versus benchmarks with looser scoring (Section 5). **Architecture scope:** every system evaluated is a single-agent ReAct loop; multi-agent decompositions (Planner-Worker, Squad architectures) are known to deliver large gains on adjacent benchmarks [16, 18] and would be a natural extension.

5.5 Future Work

Several extensions follow directly from these limitations. **Scoring:** we will add tolerance-/range-based scoring so near-equal comparisons and numeric estimates are not failed by strict binary matching, and report a pass^k reliability metric (the probability a model solves a case on *all k* trials) alongside mean accuracy, since production use depends on consistency rather than best-of-*k* behavior. **Coverage:** expanding the open- and closed-weight roster, growing the thin categories, and adding regions and indicators would raise statistical power and test geographic generalization. **Substrate:** exposing the benchmark’s other MCP primitives (resources, prompts) and evaluating subagent-orchestrated clients would broaden the agentic constructs under test. **Sustainability:** refining the energy/carbon estimate (Section 4.4) toward measured per-token footprints would make the environmental cost of deployment a first-class, comparable metric.

6 Conclusion

We introduce GeoNatureAgent Benchmark, a benchmark for evaluating AI agents on environmental geospatial analysis through structured tool calling against an open, self-hostable API. The benchmark spans three indicators across Spain and Portugal, demonstrating domain and geographic extensibility. Evaluating nine LLMs on 93 tasks across 18 categories with sixteen tools, under three temperature-1.0 seeds per model and reporting capability and per-case cost as orthogonal axes:

- (1) **Environmental geospatial tool orchestration remains open.** The capability leader (Claude Sonnet 4) achieves 60.8%±0.8%, well below the 85–97% reported on general GIS benchmarks; only two models exceed 55% and none reaches 65%.
- (2) **Open-weight models occupy the cost-accuracy Pareto frontier.** Three of the four models on the Pareto frontier

(Llama 4 Scout, Qwen3-235B, and DeepSeek V3.2) are open-weight; closed-source Gemini 2.5 Pro and the larger open-weight GPT-OSS-120B are dominated.

- (3) **The capability-cost trade-off is steep.** DeepSeek V3.2 delivers 93% of Claude’s capability (56.3% vs. 60.8%) at 11.6× lower per-case cost, making it the strongest single-model recommendation for cost-sensitive production deployment.
- (4) **Systematic failures persist.** Tool-selection variance remains the dominant inter-model spread, and parameter scale alone does not predict capability.

Future work: expanding to additional countries and indicators (e.g. NDVI, precipitation, and biodiversity indices); prompt ablation (zero-shot, CoT, few-shot); and multi-agent architecture comparison following the Planner-Worker paradigm of GeoJSON Agents [18].

GeoNatureAgent Benchmark, evaluation code, and the full MLOps pipeline are publicly available.²

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²https://github.com/gabrielireland/GeoNatureAgent_Benchmark

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A Online Resources

The GeoNatureAgent Benchmark code, evaluation harness, and self-hostable API are available at https://github.com/gabrielireland/GeoNatureAgent_Benchmark. The benchmark dataset (tasks, per-case results, and the derived environmental indicators for Spain and Portugal) is archived at Zenodo under DOI <https://doi.org/10.5281/zenodo.20450995> [7], and the evaluation software is also archived at Zenodo under DOI <https://doi.org/10.5281/zenodo.20450997> [8].